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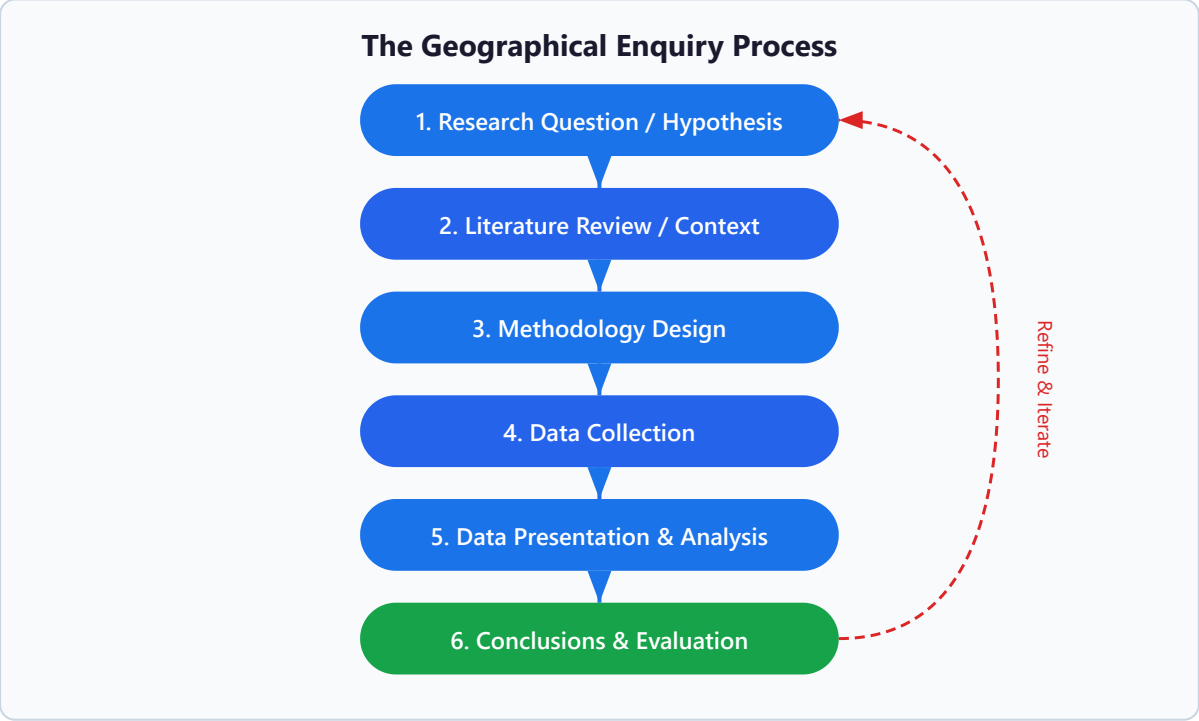
Fieldwork & Investigation (AQA Paper 3)

AQA Paper 3 assesses your ability to design, execute, and critically evaluate geographical investigations. You must demonstrate understanding of the entire enquiry process, from formulating a question through to evaluation. This section covers the methodological, statistical, and presentational skills required. You will answer questions on your own two fieldwork investigations (one physical, one human) and on an unfamiliar investigation presented in the exam.

1. The Enquiry Process

Geographical enquiry: A systematic investigation that follows a structured process to answer a geographical question or test a hypothesis, using primary and/or secondary data. The enquiry process is iterative — findings may prompt refinement of the question or methodology.



Formulating Questions and Hypotheses

A good geographical question is specific, measurable, achievable, relevant, and rooted in geographical theory. It should focus on a relationship, pattern, or process that can be investigated with available data.

- **Null hypothesis (H_0):** States there is no significant relationship, difference, or pattern. E.g. "There is no significant correlation between distance from the CBD and land values." This is the hypothesis you test statistically.
- **Alternative hypothesis (H_1):** States there IS a significant relationship, difference, or pattern. E.g. "There is a significant negative correlation between distance from the CBD and land values." This is what you expect to find based on theory.

Good questions often derive from geographical models or theories: Does the Bradshaw model apply to this river? Does the Burgess model explain land use patterns in this town? Does environmental quality decline with distance from a park?

Exam Tip: Always frame your hypothesis as a testable, two-part statement (null + alternative). Examiners reward students who clearly operationalise their variables (define exactly what they will measure and how) and state the expected direction of the relationship. Weak hypotheses are vague ("I think the river will change downstream"); strong hypotheses are specific and justified ("There is a significant positive correlation between distance downstream and channel width, because discharge increases with contributing area").

Risk Assessment and Ethical Considerations

All fieldwork investigations must include a risk assessment and address ethical considerations:

Risk Assessment	Ethical Considerations
Identify physical hazards: fast-flowing water, unstable cliffs, traffic, tides, weather	Obtain informed consent for questionnaires and interviews
Assess likelihood and severity of each hazard	Respect privacy and confidentiality of respondents
Implement mitigation: PPE, buddy system, mobile phone, adult supervision, tidal timetables	Avoid sensitive or intrusive questions

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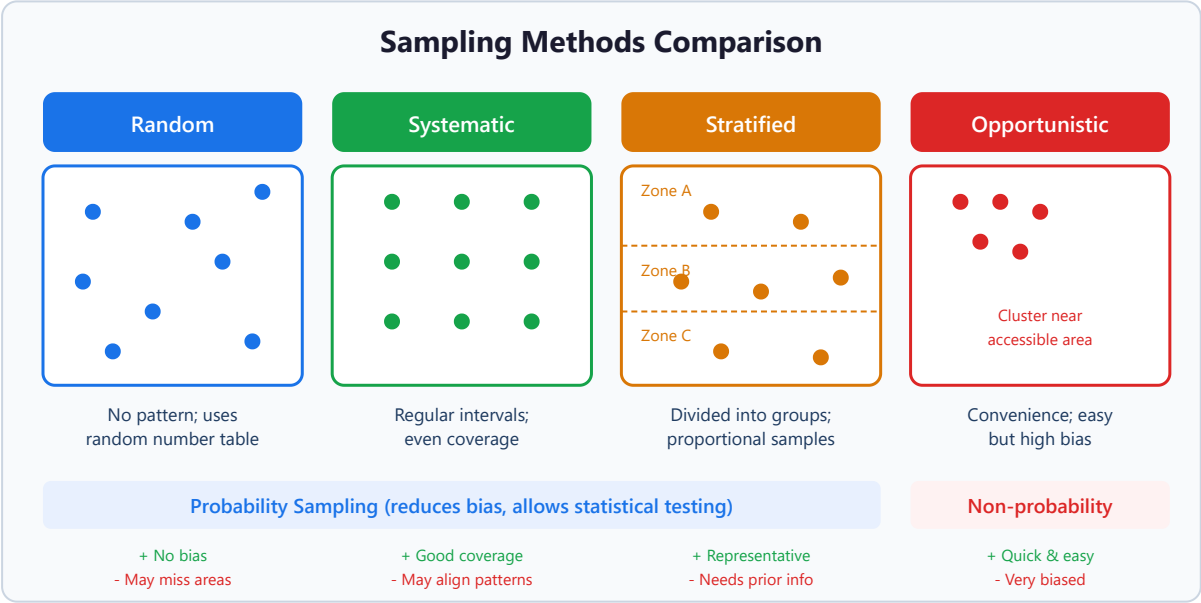
Emergency procedures: first aid kit, emergency contacts, nearest hospital	Minimise environmental impact (don't damage habitats, replace stones)
Check weather forecasts before fieldwork	Respect private property and access rights
Ensure group awareness of meeting points and return times	Present findings honestly without fabrication or distortion

2. Sampling Techniques

Sampling: The process of selecting a subset of individuals, locations, or measurements from a larger population for study, since it is usually impractical (and unnecessary) to collect data from every possible point. The choice of sampling technique directly affects the reliability, representativeness, and statistical validity of results.



Sampling Method	Description	Best Used When	Key Limitation

Random	Each member of the population has an equal chance of selection. Uses random number tables or generators to select sample points on a grid.	Study area is relatively uniform; you need statistically valid results free from researcher bias.	May leave spatial gaps; requires a sampling frame (numbered grid or list).
Systematic	Samples taken at regular intervals (e.g. every 10th house, every 50 m along a transect, every 5 minutes).	You need good spatial or temporal coverage along a transect or route; easy to implement in the field.	May coincide with periodic patterns in data (e.g. measuring at every street corner could bias results).
Stratified	Population is divided into sub-groups (strata), then random or systematic samples are taken from each in proportion to their size.	Population has distinct sub-groups that should all be represented (e.g. different land use zones, age groups, vegetation communities).	Requires prior knowledge of population composition; more complex to organise.
Quota	Researcher sets a target number for each sub-group and selects participants until each quota is filled. Non-random — researcher chooses who to approach.	Questionnaire surveys where you need specific demographic representation but lack a sampling frame.	High researcher bias (tendency to approach certain "types"); not statistically representative.
Opportunistic (convenience)	Samples collected wherever and whenever convenient. No structured approach.	Time or access constraints make other methods impractical.	Highly biased; results cannot be generalised; should only be used as a last resort with limitations acknowledged.

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Exam Tip: In Paper 3, you MUST justify your sampling technique by linking it to your specific investigation. Do not just describe the method — explain WHY it was appropriate for your study area, data type, and research question. Always acknowledge limitations and explain how they might affect your results. E.g. "We used systematic sampling at 50 m intervals along the river to ensure even spatial coverage of the study reach. A limitation is that this may miss localised features between sample points."

3. Data Collection: Primary vs Secondary

Primary data: Data collected first-hand by the researcher through direct observation, measurement, or surveys in the field. You have control over methodology, timing, and accuracy, but collection is time-consuming and limited in scope.

Secondary data: Data collected by someone else and made available for use. Includes census data (ONS), published statistics, Ordnance Survey maps, satellite imagery (Google Earth), Environment Agency flood/river data, weather station records, IMD data, historical photographs. Allows access to large datasets and historical trends, but you have no control over collection methods and data may not precisely match your needs.

Common Physical Geography Data Collection Methods

Measurement	Equipment	Method	Accuracy Considerations
Channel width	Tape measure	Stretch across channel at bankfull level, perpendicular to flow	Identify bankfull accurately; ensure tape is taut and level
Channel depth	Metre ruler or ranging pole	Measure at regular intervals across the channel; calculate mean	Take multiple readings (e.g. every 50 cm across); bed may be uneven
Flow velocity	Flow meter (impeller), or float	Flow meter: hold at 0.6 depth (represents mean velocity).	Flow meters give more accurate readings; floats measure surface

	+ stopwatch + tape measure	Float: time an orange over 10 m, repeat 3x	velocity (multiply by 0.85 for mean). Take multiple readings.
Discharge	Calculated: Q = A × V	Cross-sectional area (width × mean depth) multiplied by mean velocity	Accuracy depends on quality of width, depth, and velocity measurements
Sediment size	Callipers or ruler	Measure b-axis (intermediate axis) of 30+ pebbles at each site using random sampling (eyes closed, pick nearest stone)	Minimum 30 samples for statistical validity; Powers' Roundness Index for shape
Gradient/slope	Clinometer or ranging poles + clinometer	Sight along slope between two points of equal height; read angle	Ensure consistent measurement height; use multiple readings
Infiltration rate	Infiltration ring (metal cylinder), measuring cylinder, stopwatch	Push cylinder 5 cm into soil, add measured volume of water, time how long it takes to infiltrate	Antecedent moisture conditions affect results; repeat 3x at each site
Soil moisture	Soil moisture probe or "feel" estimation	Insert probe to consistent depth; record reading	Calibrate probe; take multiple readings per site

Common Human Geography Data Collection Methods

Method	Description	Advantages	Limitations
Questionnaires	Structured set of questions (closed and/or open-ended) administered to respondents	Standardised, comparable, can reach large samples, quantitative analysis possible	Response bias, social desirability, low response rates, question wording influences answers

Interviews	Semi-structured conversations with individuals (in-depth, qualitative)	Rich, detailed data; can follow up on interesting responses; reveals perceptions and emotions	Time-consuming, small sample, interviewer bias, difficult to generalise
Environmental Quality Index (EQI)	Scoring system rating aspects of urban/rural environments (litter, graffiti, noise, vegetation, building quality) on a numerical scale	Quantifies subjective assessments; allows comparison between sites	Subjectivity (different observers may score differently); bipolar scales need careful design
Land use mapping	Recording the function of buildings/spaces on a base map (residential, commercial, industrial, recreational)	Visual, systematic, identifies spatial patterns	Time-consuming, some uses hard to classify, may miss upper floors
Pedestrian/traffic counts	Counting people or vehicles passing a point in a fixed time period	Quantitative, easy to standardise, reveals flow patterns	Varies with time of day/week/season; one count may not be representative
Photography	Systematic photographic survey of sites	Visual evidence, can be annotated, supports other data	Subjective framing, cannot capture everything, weather/lighting affects quality

4. Data Presentation

Choose presentation methods appropriate to your data type and purpose. Each method has strengths and limitations.

Technique	Best For	Key Features

Bar charts	Comparing discrete categories (e.g. land use types, questionnaire responses)	Simple, clear, easy to read. Use for nominal/ordinal data. Bars should not touch (discrete data).
Histograms	Showing frequency distribution of continuous data (e.g. pebble sizes, flow velocity)	Bars touch (continuous data). Shows distribution shape (normal, skewed, bimodal).
Line graphs	Showing change over time or distance (e.g. discharge downstream, temperature over a day)	Trend clearly visible. Can overlay multiple datasets. Use for interval/ratio data.
Scatter graphs	Showing relationship between two variables (e.g. distance from CBD vs house price)	Reveals correlation direction and strength. Add best-fit line. Essential for Spearman's rank.
Pie charts	Showing proportions of a whole (e.g. employment sectors, land use breakdown)	Simple proportional display. Limited to 5–7 categories maximum. Avoid for small differences.
Choropleth maps	Showing spatial variations in data using shading (e.g. IMD scores, population density)	Visual, spatial patterns clear. Use graduated colour scheme. Limitations: implies uniform distribution within each area.
Isoline maps	Showing continuous spatial data (e.g. temperature, height, pollution levels)	Lines join points of equal value. Reveals gradients and patterns. E.g. contour maps, UHI maps.
Proportional symbols	Showing quantities at specific locations (e.g. city population, discharge at gauging stations)	Visual impact, good for located data. Circles or squares proportional to value.
Cross-sections / transects	Showing change across a line (e.g. river cross-section, vegetation transect, urban transect)	Reveals spatial patterns and gradients. Useful for physical geography fieldwork.

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Dispersion diagrams	Showing spread and central tendency of data	Box-and-whisker plots show median, quartiles, range, and outliers. Good for comparing datasets.

Exam Tip: When choosing presentation methods, justify your choice: "I used a scatter graph because I wanted to identify the relationship between two continuous variables (distance downstream and pebble size) and calculate Spearman's rank correlation coefficient." Show that you understand WHY a method is appropriate, not just what it looks like.

5. Statistical Analysis

Statistics help you move beyond description to identify significant patterns, relationships, and differences in your data. AQA expects you to understand and apply several key statistical techniques:

Measures of Central Tendency and Dispersion

- **Mean** — sum of values / number of values. Affected by extreme values (outliers). Best for normally distributed data.
- **Median** — middle value when ranked in order. Not affected by outliers. Better for skewed data.
- **Mode** — most frequent value. Useful for categorical data.
- **Range** — highest value minus lowest. Simple but influenced by outliers.
- **Interquartile range (IQR)** — Q3 minus Q1 (middle 50% of data). More robust measure of spread, unaffected by outliers.
- **Standard deviation** — measures average deviation from the mean. Low SD = data clustered close to mean; high SD = data spread widely. For normally distributed data, ~68% of values fall within 1 SD of the mean, ~95% within 2 SD.

Spearman's Rank Correlation Coefficient (r_s)

Spearman's Rank: A non-parametric test that measures the strength and direction of the relationship between two ranked variables. Produces a value between -1 (perfect negative correlation) and +1 (perfect positive correlation). 0 = no correlation.

$$r_s = 1 - (6\sum d^2) / (n(n^2 - 1))$$

Where d = difference between ranks for each pair, n = number of pairs. After calculating r_s, compare it to the critical value table at the chosen significance level (usually 0.05 or 95% confidence) for your sample size. If $|r_s|$ > critical value, the correlation is statistically significant and you can reject the null hypothesis.

Interpreting r_s: 0–0.2 = very weak, 0.2–0.4 = weak, 0.4–0.6 = moderate, 0.6–0.8 = strong, 0.8–1.0 = very strong. Remember: correlation does not prove causation.

Chi-Squared Test (χ^2)

Chi-Squared: A test of association between two categorical variables. It compares observed frequencies with expected frequencies (what you would expect if there were no association). Used with nominal data in contingency tables.

$$\chi^2 = \sum (O - E)^2 / E$$

Where O = observed frequency, E = expected frequency (row total × column total / grand total). Compare the calculated χ^2 with the critical value from the table, using degrees of freedom = (rows − 1) × (columns − 1). If calculated χ^2 > critical value at the 0.05 significance level, the association is statistically significant.

Mann-Whitney U Test

Mann-Whitney U: A non-parametric test comparing two independent samples to determine if there is a significant difference between them. Used when data is ordinal or not normally distributed. E.g. comparing pebble sizes at two different beach locations, or EQI scores in two different streets.

Statistical Significance and Confidence Levels

- **Significance level (p-value)** — the probability that the result occurred by chance. At p = 0.05 (5%), there is a 95% chance the result is genuine and a 5% chance it is due to random variation.
- **Rejecting H_0** — if your test statistic exceeds the critical value at your chosen significance level, you reject the null hypothesis and accept the alternative hypothesis.

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- **Degrees of freedom** — a value calculated from sample size that determines which critical value to use. For Spearman's: df = n − 2. For chi-squared: df = (r − 1)(c − 1).
- **Confidence level** — the complement of the significance level. At p = 0.05, the confidence level is 95%. At p = 0.01, it is 99%.

Exam Tip: When presenting statistical results, always: (1) state the test used and why, (2) state the calculated value, (3) state the critical value and significance level, (4) state the degrees of freedom or sample size, (5) state whether you accept or reject H_0, and (6) explain what this means in geographical terms. E.g. "Spearman's r_s = -0.87, which exceeds the critical value of 0.648 at p = 0.05 with n = 12. Therefore, I reject H_0 and conclude there is a significant negative correlation between distance downstream and pebble size. This supports the Bradshaw model prediction that attrition reduces sediment size downstream."

6. Conclusions and Evaluation

Drawing Conclusions

- Relate findings back to your original question/hypothesis — do your results support or contradict it?
- Link findings to geographical theory or models — does your data support the Bradshaw model, Burgess model, or relevant theory?
- Identify patterns, anomalies, and unexpected results — explain possible reasons for anomalies.
- Consider alternative explanations for your findings.
- Avoid overgeneralising from a small sample or limited study area.

Critical Evaluation

Evaluation is one of the highest-order skills assessed. You should critically assess:

- **Accuracy** — how close were measurements to the true value? Equipment limitations (e.g. flow meters have ±5% accuracy), human error (reading scales incorrectly), environmental conditions (wind, rain affecting measurements).
- **Reliability** — would repeating the investigation produce similar results? Did you take enough repeat measurements? Were conditions consistent across all sites?

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- **Representativeness** — can your findings be generalised to a wider area or population? Was your sample size large enough? Was your sampling strategy appropriate? Did you avoid spatial/temporal bias?
- **Validity** — did you actually measure what you intended to measure? Did your methodology address your research question?
- **Data limitations** — gaps in data, weather constraints, time restrictions, access issues, equipment availability, seasonal timing.
- **Improvements** — what would you do differently? Larger sample size, more sites, different time of year, better equipment, additional data sources, longer study period.
- **Extensions** — what further research questions arise from your findings? How could you build on this investigation?

Exam Tip: The evaluation section is where many students lose marks by being too superficial. Do not just write "I would increase my sample size." Instead, explain: "My sample size of 10 sites along a 3 km stretch was insufficient to capture the full downstream variation. Increasing to 20 sites at 150 m intervals would provide more robust data for statistical analysis (Spearman's rank requires a minimum of 10 pairs for meaningful results, and our n=10 was at this threshold). Additionally, collecting data in both summer and winter would reveal seasonal variations in discharge and velocity, which our single-day fieldwork could not capture."

Exam Tip: For Paper 3 Section B (the unfamiliar fieldwork investigation), you will be given a scenario and asked to critically evaluate someone else's methodology, data, and conclusions. Practice reading data presentation techniques, identifying strengths and weaknesses of methodologies, and suggesting improvements. The key skills are: interpreting graphs/maps/statistics you have not seen before, identifying sampling bias, assessing whether conclusions are supported by the data, and suggesting alternative methodologies.

Exam Tip: AQA expects you to demonstrate that your fieldwork is connected to the wider specification. For a river investigation, link to water cycle processes (infiltration, runoff, throughflow) and the drainage basin system. For an urban investigation, link to place identity, inequality, or sustainability. This synoptic connection lifts your evaluation from competent to excellent.

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7. Spearman's Rank: Full Worked Example

Below is a complete worked example of Spearman's Rank Correlation, the most commonly required statistical test in AQA Geography Paper 3.

Research Question and Data

Research question: "Is there a significant relationship between distance downstream from the source and mean pebble size at a river in the Lake District?"

Null hypothesis (H_0): There is no significant correlation between distance downstream and mean pebble size.

Alternative hypothesis (H_1): There is a significant negative correlation between distance downstream and mean pebble size (pebbles become smaller with distance as attrition and abrasion reduce size).

Site	Distance from source (km)	Mean pebble b-axis (mm)	Rank: Distance (R_1)	Rank: Pebble size (R_2)	Difference ($d = R_1 - R_2$)	d^2
1	0.5	87	1	10	−9	81
2	1.2	74	2	9	−7	49
3	2.1	68	3	8	−5	25
4	3.4	61	4	7	−3	9
5	4.7	55	5	6	−1	1
6	5.9	48	6	5	1	1
7	7.2	42	7	4	3	9
8	8.8	36	8	3	5	25
9	10.1	29	9	2	7	49
10	12.3	21	10	1	9	81

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$\sum d^2 =$	**330**

Calculation

$$r_s = 1 - (6 \times \sum d^2) / (n(n^2 - 1))$$

$$r_s = 1 - (6 \times 330) / (10 \times (100 - 1))$$

$$r_s = 1 - 1980 / (10 \times 99)$$

$$r_s = 1 - 1980 / 990$$

$$r_s = 1 - 2.0 = \mathbf{-1.0}$$

Interpreting the Result

$r_s = -1.0$ is a perfect negative correlation (in this simplified example). In real fieldwork, r_s values between −0.7 and −0.9 are typical for downstream pebble-size studies, indicating a strong negative correlation with some variation.

Critical value check: For n = 10 at p = 0.05 (95% confidence level), the critical value from Spearman's Rank table is **0.648**. Since |−1.0| = 1.0 > 0.648, we **reject H_0**.

Conclusion: There is a statistically significant negative correlation between distance downstream and mean pebble size ($r_s = -1.0$, n = 10, $p < 0.05$). We accept H_1. This supports the theoretical prediction that attrition and abrasion reduce pebble size with increasing distance downstream, consistent with the Bradshaw model's prediction of decreasing sediment particle size downstream.

Exam Tip: Note two important conventions when ranking: (1) When ranking data for Spearman's, rank 1 is assigned to the *smallest* value if you are correlating two variables that should both increase (or both decrease) together. Be consistent — decide before you start whether rank 1 = smallest or largest, and apply this consistently to BOTH variables. (2) For *tied values*: if two values are equal,

assign them the average of the ranks they would have occupied (e.g. if values ranked 3rd and 4th are equal, both get rank 3.5). This must be done correctly or your r_s calculation will be wrong.

8. Chi-Squared: Worked Example

Research Question and Observed Data

Research question: "Is there a significant association between proximity to a city park and residents' reported use of outdoor green space for exercise?"

H_0: There is no significant association between proximity to the park and frequency of outdoor exercise.

H_1: There is a significant association between proximity to the park and frequency of outdoor exercise.

Observed frequencies (from questionnaire survey, n = 120):

	Exercises outdoors 3+ times/week	Exercises outdoors fewer than 3 times/week	Row Total
Lives within 300 m of park	42	18	60
Lives more than 300 m from park	24	36	60
Column Total	66	54	120

Step 1: Calculate Expected Frequencies

Expected frequency (E) for each cell = (Row total × Column total) / Grand total

- Near park, exercises frequently: (60 × 66) / 120 = 33
- Near park, exercises infrequently: (60 × 54) / 120 = 27
- Far from park, exercises frequently: (60 × 66) / 120 = 33
- Far from park, exercises infrequently: (60 × 54) / 120 = 27

Step 2: Calculate Chi-Squared

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$$\chi^2 = \sum (O - E)^2 / E$$

Cell	O (Observed)	E (Expected)	(O − E)	$(O - E)^2$	$(O - E)^2 / E$
Near, frequent	42	33	9	81	2.45
Near, infrequent	18	27	−9	81	3.00
Far, frequent	24	33	−9	81	2.45
Far, infrequent	36	27	9	81	3.00
				χ^2 =	**10.91**

Step 3: Compare to Critical Value

Degrees of freedom = (rows − 1) × (columns − 1) = (2−1) × (2−1) = 1

Critical value at p = 0.05, df = 1: **3.84**

Critical value at p = 0.01, df = 1: **6.63**

Our calculated χ^2 = 10.91 > 6.63 (p = 0.01 critical value). Therefore we **reject H_0** at the 99% confidence level.

Conclusion: There is a highly significant association between proximity to the park and frequency of outdoor exercise (χ^2 = 10.91, df = 1, p < 0.01). We accept H_1. People living within 300 m of the park are more likely to exercise outdoors 3+ times per week. This is consistent with the concept that accessible urban green space promotes physical activity and health.

Exam Tip: Chi-squared has strict requirements that examiners test: (1) data must be in *frequencies* (counts), not percentages or proportions — convert percentages back to counts if the total sample size is given; (2) no expected frequency should be less than 5 — if any cell has E < 5, you must either collapse categories or use an alternative test; (3) observations must be *independent* — each person/location should only appear in one cell. Violating these conditions invalidates the test. In your evaluation, mention any cells that approached the E < 5 threshold as a limitation.

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9. Hypothesis Testing Framework

AQA Geography Paper 3 requires a rigorous approach to hypothesis testing. The following framework applies to all statistical tests:

Step	Action	Example
1	State the research question clearly	"Does land use diversity decrease with distance from the CBD?"
2	State H_0 (null hypothesis) — always states NO significant relationship/difference	"There is no significant correlation between distance from the CBD and land use diversity index score."
3	State H_1 (alternative hypothesis) — states the expected pattern, justified by theory	"There is a significant negative correlation between distance from the CBD and land use diversity, because competing land values near the CBD favour commercial monoculture." (based on Burgess model)
4	Identify the appropriate statistical test and justify its selection	"Spearman's Rank is appropriate because both variables are ordinal/continuous, the relationship is expected to be monotonic (consistently increasing or decreasing), and the sample is too small for parametric tests (n = 12)."
5	State the significance level and justify it	"I will use p = 0.05 (95% confidence level), which is standard for geographical fieldwork. A 5% probability of random results being accepted as significant is acceptable for this exploratory investigation."
6	Collect data using appropriate sampling method	Systematic sampling at 200 m intervals along 5 transect routes from CBD to urban fringe.
7	Calculate the test statistic	Calculate r_s using the formula, showing working.
8	Compare calculated value to critical value from table	$r_s = -0.73$; critical value for n = 12 at p = 0.05 is 0.591. Since 0.73 > 0.591, the result is significant.

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9	State conclusion: accept or reject H_0	"I reject H_0. There is a significant negative correlation between distance from the CBD and land use diversity ($r_s = -0.73$, $n = 12$, $p < 0.05$)."
10	Interpret the result geographically — link to theory	"This supports the Burgess concentric zone model's prediction that land use near the CBD is dominated by commercial uses (low diversity) while the residential zone contains a greater mix of uses. However, two anomalous sites with high commercial diversity at 1.8 km distance may reflect out-of-town retail developments."
11	Evaluate: what are the limitations of this conclusion?	"The sample size ($n = 12$) is near the minimum for Spearman's Rank; a larger sample would increase confidence. The transect routes chosen may not be representative of all directions from the CBD. The land use diversity index is a subjective measurement instrument and different observers may score the same street differently."

10. Primary vs Secondary Data: Evaluation Framework

Criterion	Primary Data	Secondary Data
Definition	Collected first-hand by the researcher through direct observation, measurement, or survey	Collected by another party and made available for use by others
Control over methodology	Full control: researcher decides when, where, how, and what to measure. Can be tailored precisely to the research question.	No control: methodology was determined by the original collector. May not precisely match your research question.
Accuracy	Accuracy depends on researcher's skill, equipment quality, and conditions at time of collection. Errors can be identified and controlled.	Accuracy depends on the original collector's methodology. Errors may be unknown or undisclosed. Errors may be systematic across the entire dataset.

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Currency/timeliness	Can be collected at the exact time needed (e.g. during a flood event, immediately after a storm).	May be outdated: census data is 10 years old at collection time; Environment Agency river data may have gaps. Historical data allows time-series analysis that primary data cannot provide.
Scale and coverage	Limited to what one researcher (or group) can physically collect in the time available. Typically small sample sizes and limited spatial coverage.	Can cover large areas and long time periods impossible for one researcher: ONS census covers entire population; EA river gauging stations record 15-minute intervals continuously.
Cost and access	Free to collect but time-intensive. Access to some sites may be restricted.	Much secondary data is free and publicly accessible (ONS, EA, OS, police.uk, IMD). Some datasets require purchase or institutional access.
Bias	Researcher bias in collection (questionnaire wording, sampling location choices, measurement technique). Can be designed to minimise bias.	Bias embedded in the original collection: government statistics may reflect political priorities; media representations are inherently selective; company data may be partial.
Examples (physical)	River width/depth/velocity measurements; pebble size; infiltration rates; vegetation quadrats; soil pH; weather observations	Environment Agency gauging station records; Met Office weather data; BGS geological maps; OS maps; satellite NDVI imagery; EA flood risk maps
Examples (human)	Questionnaires; interviews; EQI surveys; pedestrian counts; land use mapping; photographic surveys; traffic counts	ONS Census; IMD data; Land Registry house prices; crime statistics; NHS health data; Google Street View; social media analytics; historical maps and photographs

Evaluating Secondary Data Sources: Key Questions

- **Who collected it, and why?** A government report on immigration may undercount irregular migrants; a company's environmental performance data may be selectively positive. Identify the collector and their purpose.
- **When was it collected?** The 2021 UK Census is the most recent but already 3 years old; the IMD was last updated in 2019 (using 2017–18 data). Areas may have changed substantially.
- **What methodology was used?** Census data is comprehensive but self-reported (people may misreport). Police crime statistics undercount unreported crimes (typically only 40% of crimes are reported).
- **What are the spatial and temporal units?** IMD data is at LSOA level (~1,500 people); averaging across larger areas may mask significant intra-area variation.
- **Are there gaps or known biases?** The 2021 Census estimated 10% non-response in some London boroughs; LA-level data may mask significant neighbourhood variation.
- **Has it been published in a peer-reviewed or quality-assured context?** ONS, EA, and academic journals are more reliable than tabloid news coverage or advocacy group statistics.

Exam Tip: In the Paper 3 unfamiliar investigation section, you will often be asked to evaluate secondary data sources. Apply the questions above systematically. Do not simply say "secondary data is less reliable than primary data" — this is too vague. Instead, identify SPECIFIC limitations of the specific secondary source presented (e.g. "The crime statistics used may undercount actual crime because many crimes go unreported; the researcher has no way of knowing the unreported crime rate in this area").

11. Data Presentation Techniques: Extended Analysis

Choosing the Right Presentation Method

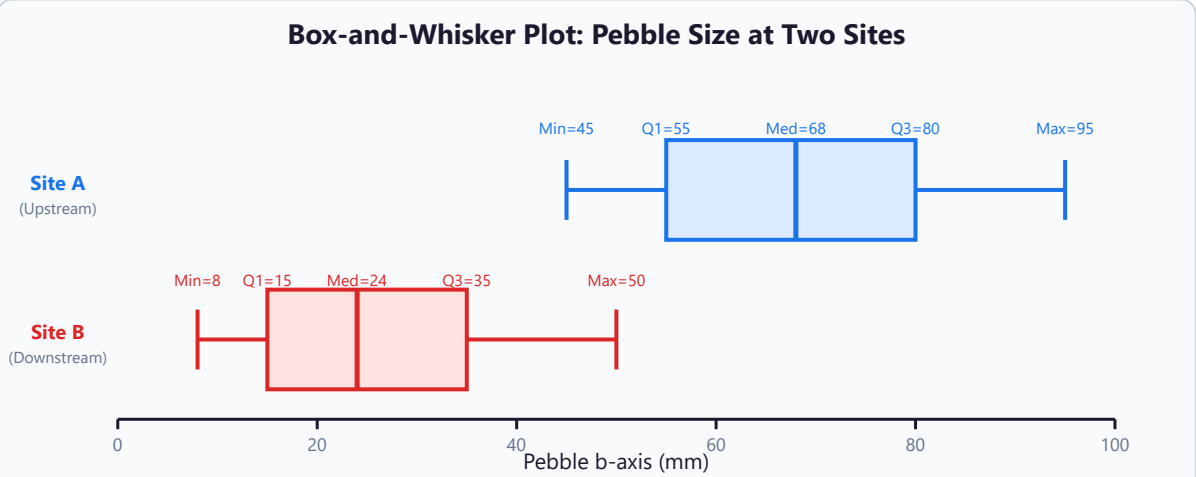
The choice of presentation technique must be justified in terms of: (1) data type (nominal, ordinal, interval, ratio); (2) what you want to show (trend, distribution, relationship, spatial pattern, proportion); and (3) the audience and purpose.

Data Type	Definition	Examples in Geography	Appropriate techniques

Nominal	Categories with no numerical order or ranking	Land use type (residential, commercial, industrial); vegetation community; rock type	Bar chart, pie chart, choropleth (categories), pictogram
Ordinal	Categories that can be ranked but intervals are not equal	Beaufort wind scale; Richter scale; EQI scores; IMD quintile ranking	Bar chart, ranked data table, Spearman's Rank, percentile charts
Interval	Numerical data with equal intervals but no true zero	Temperature in °C; years (AD/BC); pH scale	Line graph, scatter graph, histogram, box-and-whisker
Ratio	Numerical data with equal intervals AND a true zero	Rainfall (mm); discharge (m^3/s); population; distance; pebble size	All techniques; allows ratios (twice as much) to be meaningful

Box-and-Whisker Plots

Box-and-whisker plots (box plots) are particularly useful in geographical fieldwork for comparing the distribution and spread of two or more datasets. They display five summary statistics: minimum, lower quartile (Q1), median (Q2), upper quartile (Q3), and maximum.



Site A (upstream) has consistently larger pebbles than Site B (downstream). No overlap in IQR.

Reading this box-and-whisker plot:

- **Site A** (upstream) has a median pebble size of 68 mm and IQR of 25 mm (Q1=55, Q3=80), indicating a wide spread of pebble sizes.

- **Site B** (downstream) has a median pebble size of 24 mm and IQR of 20 mm (Q1=15, Q3=35), indicating smaller pebbles with less variation.
- The fact that the boxes (IQR) **do not overlap** is strong visual evidence of a significant difference in distributions between the two sites, consistent with downstream fining.
- A Mann-Whitney U test could be used to formally test whether this difference is statistically significant.

12. Evaluating Your Fieldwork Investigation: A Framework

The AREA Framework for Evaluation

Element	Questions to ask	Example of high-quality evaluation
Accuracy	How close were your measurements to the true value? What sources of measurement error existed?	"Flow velocity measurements using a floating orange are known to overestimate mean velocity as they measure surface velocity. We multiplied all values by 0.85 to correct for this, but this correction factor is itself an approximation and may introduce error of ±10%."
Representativeness	Does your sample represent the population from which it was drawn? What spatial or temporal biases exist?	"We conducted all fieldwork on a single weekday in October. Consumer behaviour and pedestrian counts in our CBD study would differ significantly on a Saturday, during school holidays, or in December. Results may not represent the typical weekday pattern either, as we only collected data between 10am and 2pm."
Evidence base	Is your sample size large enough to draw reliable conclusions? Do you have enough data points for statistical analysis?	"Our Spearman's Rank analysis used n=8 sites, which is at the lower boundary of reliability for this test. The critical value at n=8 (p=0.05) is 0.738, meaning we needed a very strong correlation to achieve significance. Increasing to n=15+ sites would lower the required threshold and increase the test's power."

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Alternative explanations	Could factors other than your hypothesised relationship explain your findings?	"The negative correlation we found between distance from the CBD and Environmental Quality Index scores could reflect distance from the CBD, OR it could reflect the age of housing stock (which is older nearer the CBD), proximity to traffic (main roads radiate from the CBD), or socioeconomic status (which also varies with distance in the Burgess model). We cannot isolate the specific causal variable from our data alone."

Improvements and Extensions: How to Write Them Well

Examiners reward *specific, justified* improvements. Avoid generic statements. For each improvement, explain:

1. **What** you would change
2. **How** specifically (the methodology)
3. **Why** this would improve the investigation (the limitation it addresses)
4. **What** you would expect to find differently (the predicted impact on results)

Example of a weak improvement statement: "I would collect more data."

Example of a strong improvement statement: "I would increase the number of transect sites from 8 to 20, spaced at 500 m intervals along the full 10 km study reach, rather than our current irregular spacing. This would improve the spatial representativeness of our dataset and provide a sufficient sample size (n=20) to detect moderate correlations in Spearman's Rank (critical value at n=20, p=0.05 is 0.450, compared to 0.738 at n=8). Currently, our irregular spacing means that a 3 km section of the middle reach is represented by only 2 data points, which may be insufficient to capture the full variation in that section."

Exam Tip: Paper 3 Section A (your own fieldwork) and Section B (unfamiliar investigation) both reward the same evaluation skills. For Section B, you are given an investigation you have not seen before and asked to evaluate its methodology. The skills are identical: identify what the researcher did, evaluate its strengths and limitations, suggest improvements, and interpret whether the conclusions are supported by the data. Practise by evaluating past-paper scenarios as well as your own fieldwork.

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Exam Tip: AQA Paper 3 is 2 hours 30 minutes for 120 marks. Section A (your own fieldwork) = 48 marks; Section B (unfamiliar investigation) = 36 marks; Section C (synoptic essay) = 36 marks. Time management is critical. For each mark, budget approximately 1.25 minutes. The synoptic essay (36 marks) should take ~45 minutes. Practise writing concise, evidence-based answers under timed conditions.

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